1.1 $\sqrt{2}$ 는 유리수인가?

$\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$

자연수 정수 유리수 실수 복소수

Ordered Field

Field.

사칙연산

Field : $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$

순서

Ordered Field : $\mathbb{Q}$, $\mathbb{R}$

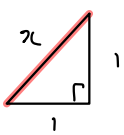
완비성

complete ordered Field $\Longleftrightarrow$ $\mathbb{R}$

(정의)

"기하학은 완비성을 가진 수체계를 요구한다."

if $x^2 = 2$ then $x \notin \mathbb{Q}$



(Pythagoras)

길이 $\longleftrightarrow$ 수

(기하학) (산술학)

$\because$ suppose $(p/q)^2 = 2$

- p is even : $p^2 = 2q^2$
- q is even : $2r^2 = q^2$ (let $p = 2r$)

Axiom of Field

$a+b = b+a$

$(a+b)+c = a+(b+c)$

$a+0 = a$

$a+(-a) = 0$

$a \cdot b = b \cdot a$

$(a \cdot b) \cdot c = a \cdot (b \cdot c)$

$a \cdot 1 = a$ $(1 \neq 0)$

$a \cdot (a^{-1}) = 1$ $(a \neq 0)$

$a \cdot (b+c) = a \cdot b + a \cdot c$

Axiom of Ordered Field

$a \leq b$ then

① $a+c \leq b+c$

② $a \cdot c \leq b \cdot c$ $(0 \leq c)$

$\mathbb{Q}$ is dense in itself

$\because$ $\mathbb{Q}$ is ordered field

$$\underbrace{\frac{a+a}{2}}_{=a} < \frac{a+b}{2} < \underbrace{\frac{b+b}{2}}_{=b}$$

1.2 수학 기호.

Set Theory

- set, element
- $x \in A$, $x \notin A$
- $A \cap B$, $A \cup B$
- A^c (complement)
- $(A \cap B)^c = A^c \cup B^c$

  $(A \cup B)^c = A^c \cap B^c$ (De Morgan's law)
- $\emptyset$
- disjoint $E \cap S = \emptyset$
- $A \subseteq B$ (inclusion, subset), $A = B$

$f : \underbrace{A}_{\text{domain}} \longrightarrow \underbrace{B}_{\text{codomain}}$, $f(A) := \underbrace{\{ f(x) : x \in A \}}_{\text{range}}$

Dirichlet Function

$g : \mathbb{R} \to \{0, 1\}$

$$g(x) = \begin{cases} 1 & (x \in \mathbb{Q}) \\ 0 & (x \notin \mathbb{Q}) \end{cases}$$

Absolute Value Function

$$|x| = \begin{cases} x & (x \geq 0) \\ x & (x < 0) \end{cases}$$

- multiplication : $|ab| = |a||b|$
- triangle inequality : $|a+b| \leq |a| + |b|$
- metric property : $|a-b| \leq |a-c| + |c-b|$

$a = b \iff |a-b| < \varepsilon \quad (\forall \varepsilon > 0)$

$\because$ ($\Leftarrow$) proof via contradiction

if $a \neq b$

let $\varepsilon_0 = |a-b| > 0$

Induction

if ① $1 \in S$

② $n \in S \Rightarrow (n+1) \in S$

then $S = \mathbb{N}$

<compact함의 의미>

- topological space X is compact if $\forall X$'s open cover has finite subcover

- $X \subseteq \mathbb{R}^n$ (Euclidean Space) is compact iff $\underbrace{\text{closed}}$ & bounded

  극한값이 도망치지 않는다.

Cantor's Intersection Thm.

- let X be topological space (i.e $\mathbb{R}$)

- suppose $\{K_n\}$ is a sequence of subsets of X

$$\begin{cases} K_n \neq \emptyset \\ K_n \text{ is compact} \\ K_1 \supseteq K_2 \supseteq K_3 \cdots \end{cases} \quad \text{then} \quad \bigcap_{n=1}^{\infty} K_n \neq \emptyset$$

---

compact하다는 것 : " 아무리 쥐어짜도 (무한히 축소), 극한으로 몰아넣어도,

그 대상 (원소)이 밖으로 새어나가 (Not closed)

멀리 도망치지 못하게 (Not Bounded)

꽉 붙잡아두는 성질 "

$\mathbb{N}$은 compact하지 않다.

이유 ① Not Bounded

이유 ② $\mathbb{N} = \{1, 2, 3, 4, 5, \cdots\}$ then $\bigcup_{n \in \mathbb{N}} A_n = A_1$

A_1

A_2

A_3

$\vdots$

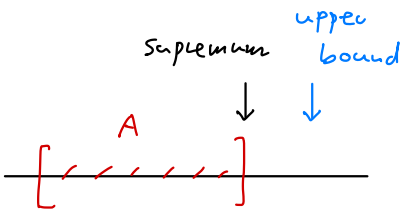
$\bigcap_{n \in \mathbb{N}} A_n = \emptyset$ $\because$ $\underbrace{n \notin A_{n+1}}$

• 아르키메데스 성질

$A_1 \supseteq A_2 \supseteq A_3 \cdots$

1.3 완비성 공리

"상계가 존재한다면 상한 (최소상계)이 존재한다"



"bounded above"

- $A \subseteq \mathbb{R}$ is bounded above if $\underbrace{\forall a \in A \quad a \le b}_{\text{bounded above}}$ $\underbrace{(\exists b \in \mathbb{R})}_{\text{upper bound}}$

$A \neq \emptyset$

- $\underbrace{\text{then}}_{\text{completeness of } \mathbb{R}}$ $\underbrace{\exists s \in \mathbb{R}}_{\text{supremum}}$ s.t. ① s is upper bound

② $s \le \underbrace{\forall b}_{\text{every upper bound}}$

---

supremum is unique $\because$ $\mathbb{R}$ is $\underbrace{\text{ordered field}}_{s_1 < s_2}$

---

a_0 is Maximum of A

iff $\begin{cases} a_0 \in A \\ a_0 \text{ is } \underbrace{\text{supremum}}_{\forall a \in A \quad a \le a_0} \end{cases}$

Note. $A := \{\frac{1}{n} : n \in \mathbb{N}\}$

then $\underbrace{\inf A = 0}_{\text{아르키메데스 성질}} \notin \underbrace{A}_{\nexists \text{Min}}$

$\mathbb{Q}$ is not complete $\because$ $\underbrace{s}_{\text{상한 후보}}^{\mathbb{Q}} < \underbrace{\xi}_{\mathbb{Q} \text{ dense in } \mathbb{R}}^{\mathbb{Q}} < \underbrace{\sqrt{2}}_{\mathbb{R}\text{에서의 상한. (이후에 증명)}}^{\mathbb{Q}^c}$.

(b/c) $S = \{r \in \mathbb{Q} : r^2 < 2\}$ then $\nexists \sup S$

$\underbrace{\sup(c+A)}_{\sup\{c+a : a \in A\}} = c + \sup(A)$ $\because$ $\forall a \le \sup(A) \le \sup(c+A) - c$

$\underbrace{\le}_{c+A\text{의 상계}}$ $\underbrace{\le}_{c+A\text{의 "최소" 상계}}$

$s = \sup A \iff s - \varepsilon < a \quad (\forall \varepsilon > 0 \quad \exists a \in A)$

"상한" "s보다 작으면 상계 아님"

suppose s is upper bound

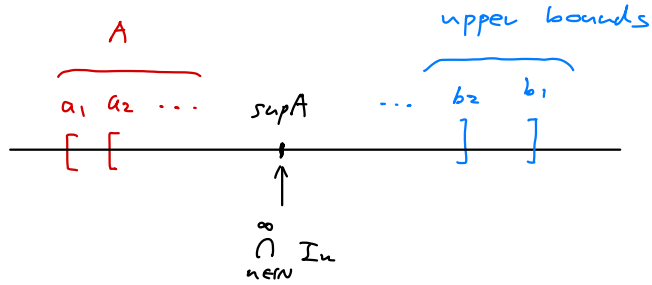
1.4 완비성 공리의 따름정리

축소구간정리.

let $\begin{cases} I_n = [a_n, b_n] \\ I_1 \supseteq I_2 \supseteq I_3 \cdots \end{cases}$

then $\bigcap_{n\in\mathbb{N}}^{\infty} I_n \neq \emptyset$

pf.



아르키메데스 성질 "$\mathbb{N}$은 위로 유계가 아니다."

(무한대) $\underbrace{x}_{\forall x\in\mathbb{R}} < \underbrace{n}_{\exists n\in\mathbb{N}}$

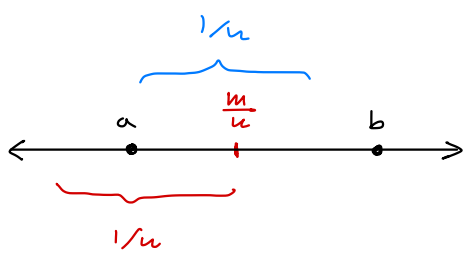
pf. $\sup\mathbb{N} - 1 < n$

$\sup\mathbb{N} < n+1$

(무한소) $0 < \underbrace{\frac{1}{n}}_{\exists n\in\mathbb{N}} < \underbrace{\varepsilon}_{\forall\varepsilon\in\mathbb{R}}$

$\mathbb{Q}$ is dense in $\mathbb{R}$

pf.



Step ① $a \le a + \underbrace{\frac{1}{n}}_{\text{매우 작음.}} < b$

Step ② $\frac{m-1}{n} \le a < \underbrace{\frac{m}{n}}_{\text{a에 착 달라붙음.}}$

then $a < \frac{m}{n} < b$

$\underbrace{\sqrt{2}}_{\text{존재한다}} = \underbrace{\sup\{t\in\mathbb{R} : t^2 < 2\}}_{=:\alpha}$

pf. otherwise if $\alpha^2 < 2$

$\left(\alpha + \frac{1}{n}\right)^2 < \alpha^2 + \frac{2\alpha+1}{n} \underbrace{< 2}_{\text{if } \frac{1}{n} < \frac{2-\alpha^2}{2\alpha+1}}$

1.5 기수

- one-to-one (단사), onto (전사), bijection (전단사)

- $A \sim B$ (cardinality) iff bijection $A \to B$ exists

- countable

(even) 2, 4, 6, 8, 10 ...

$\mathbb{N}$ 1, 2, 3, 4, 5 ...

$\mathbb{Z}$ 0, -1, 1, -2, 2 ...

uncountable

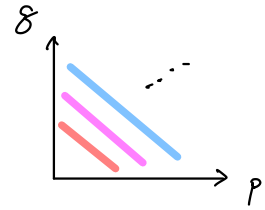
$(-\frac{\pi}{2}, \frac{\pi}{2}) \sim \mathbb{R}$

$f(x) = \tan x$

$\mathbb{Q}$ countable

$\because$

$$A_n := \left\{ \pm \frac{p}{q} : \begin{array}{l} p \cdot q \in \mathbb{N} \\ p+q = n \\ p/q \text{ 기약분수} \end{array} \right\}$$



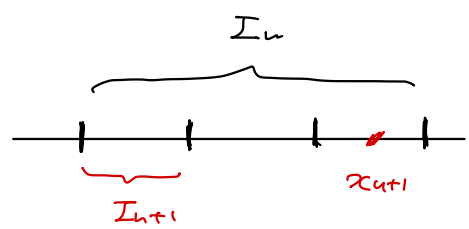
$\mathbb{R}$ uncountable

$\because$ $\mathbb{R} \Rightarrow x_1, x_2, x_3, x_4 \cdots$

closed intervals $\Rightarrow I_1 \supseteq I_2 \supseteq I_3 \supseteq I_4 \cdots$

therefore $\underbrace{\bigcap_{n \in \mathbb{N}}^{\infty} I_n = \emptyset}_{\text{축소구간성질 파괴}}$ $\forall n \in \mathbb{N}$ $x_n \notin I_n$

만드는 법 : 3등분한다.



$\underbrace{\mathbb{N}}_{\text{가산 집합}}$ 의 $\underbrace{\text{subset}}_{\text{at most countable}}$은 $\underbrace{\begin{cases} \text{finite} \\ \sim \mathbb{N} \end{cases}}_{\text{smallest inf set } (S_0)}$

$\because$ Suppose $S \subseteq \mathbb{N}$ is infinite

bijection $n \longmapsto s_n$

$s_1 = \min(S)$

$s_2 = \min(S \setminus \{s_1\})$

...

Well-ordering Principle.

[$\mathbb{N}$의 부분집합 ($\neq \emptyset$)은 최솟값을 가진다.]

---

if $\forall n \in \mathbb{N}$ $A_n \sim \mathbb{N}$

then $\bigcup_{n=1}^{\infty} A_n \sim \mathbb{N}$

$\because$

$A_1 = \{a_{11}, a_{12}, a_{13} \cdots\}$

$A_2 = \{a_{21}, a_{22}, a_{23} \cdots\}$

$A_3 = \{a_{31}, a_{32}, a_{33} \cdots\}$

1.6 한호어 정리

$(0,1)$ uncountable

$\ulcorner\because$

$\mathbb{N}$		$(0,1)$
1	$\longleftrightarrow$	$0.a_{11}\ a_{12}\ a_{13}$
2	$\longleftrightarrow$	$0.a_{21}\ a_{22}\ a_{23}$
3	$\longleftrightarrow$	$0.a_{31}\ a_{32}\ a_{33}$
		$\vdots$

mismatch!

$0.b_1\ b_2\ b_3$

Note.

$(0,1) \sim \underbrace{\{0,1\}^{\mathbb{N}}}_{\text{2진법 소수점 표현법}}$

$\underbrace{P(A)}_{\text{power set}} = \left\{ \emptyset,\ \{1\},\ \{1,2\},\ \overbrace{\{1,2,3\}}^{A} \atop \{2\},\ \{2,3\} \atop \{3\},\ \{3,1\} \right\}$ then $|P(A)| = 2^{|A|} = 2^3$

Note.

$P(\mathbb{N}) \sim \underbrace{\{0,1\}^{\mathbb{N}}}_{\text{집합에 성분이 있냐 없냐 결정}}$

Cantor's Thm

for any set A

$\nexists\ A \xrightarrow{\text{onto}} P(A)$

$\ulcorner\because$

suppose $\exists x \in A \xrightarrow[\text{onto}]{f} \{a \in A : a \notin f(a)\} \in P(A)$

A	A
$f(x) = \{a \in A : a \notin f(a)\}$	$f(x) = \{a \in A : a \notin f(a)\}$
$x \in f(x)$	$x \notin f(x)$

Note

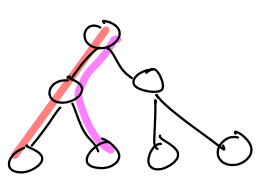
$\boxed{P(\mathbb{N}) \text{ is uncountable}}$

---

anti-chain ① $A \subseteq P(B)$

② A의 어떤 원소는 A의 다른 원소의 부분집합 아님

$P(\mathbb{N})$ contains uncountable anti-chain



(i) nodes $\sim \mathbb{N} \Longrightarrow 1+2+4+8+\cdots$ (이진수 유한소수) count

(ii) paths $\sim \mathbb{R} \Longrightarrow 2\times2\times2\times2\times\cdots$ (이진수 무한소수)

(iii) paths are anti-chain $\ulcorner\because$ $\text{path}_i \not\subseteq \text{path}_j$

1.7 카디널리티

Cantor's Thm

$$\mathbb{N} \sim \mathbb{Z} \sim \mathbb{Q}$$
$$P(\mathbb{N}) \sim \mathbb{R} \sim (0,1)$$
$$P(P(\mathbb{N}))$$
$$P(P(P(\mathbb{N})))$$

---

cardinal number

순서 : $\begin{cases} \text{card } X \le \text{card } Y & (X \xrightarrow{\text{단사}} Y) \\ \text{card } X < \text{card } Y & (X \overset{\text{단사}}{\underset{\not\longleftarrow}{\longrightarrow}} Y) \end{cases}$

칸토어 정리 : $\text{card } X < \text{card } P(X)$

가정 은 집합 U는 없다 ∵ $\text{card } U < \text{card } P(U)$

Schröder - Bernstein Thm.

if $\exists\, A \xrightarrow{\text{단사}} B$

$\exists\, B \xrightarrow{\text{단사}} A$

then $\exists\, A \xrightarrow{\text{전단사}} B$

∵ 임의의 $x \in A$에 대해 역함수 (f^{-1}, g^{-1})을 정의한다.

Search

$x_1 \to y_1 \to x_2 \to y_2 \to x \to \cdots$

$y_1 \leftrightarrow x_1 \to y_2 \leftrightarrow x \to \cdots$

$y_1 \to x_1 \to y_2 \to x$